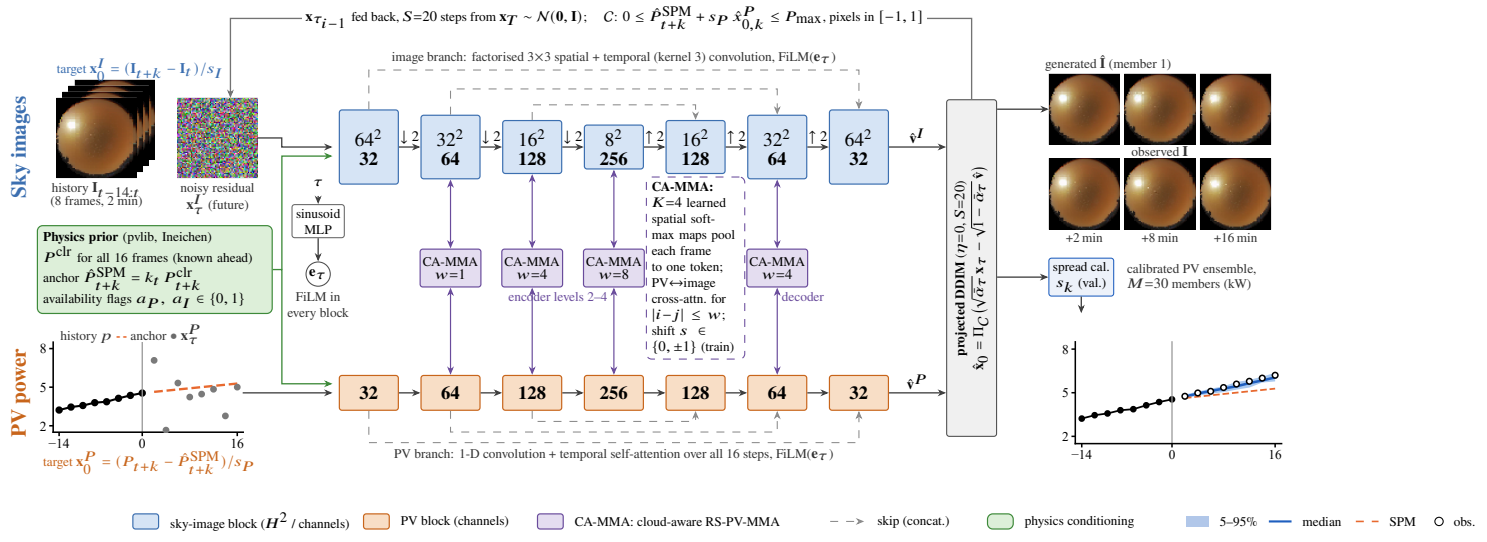

Sky images
PV power
target $\mathbf{x}_0^I = (\mathbf{I}_{t+k} - \mathbf{I}_t)/s_I$
history $\mathbf{I}_{t-14:t}$ (8 frames, 2 min)
noisy residual $\mathbf{x}_\tau^I$ (future)
$\mathbf{x}_{\tau_{i-1}}$ fed back, S=20 steps from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$; C: $0 \le \hat{P}^{\mathrm{SPM}}_{t+k} + s_P\,\hat{x}^P_{0,k} \le P_{\max}$, pixels in $[-1, 1]$
image branch: factorised 3×3 spatial + temporal (kernel 3) convolution, FiLM($\mathbf{e}_\tau$)
64^2 32
32^2 64
16^2 128
8^2 256
16^2 128
32^2 64
64^2 32
$\hat{\mathbf{v}}^I$
τ
sinusoid MLP
$\mathbf{e}_\tau$
FiLM in every block
CA-MMA w=1
CA-MMA w=4
CA-MMA w=8
CA-MMA w=4
encoder levels 2–4
decoder
CA-MMA: K=4 learned spatial soft-max maps pool each frame to one token; PV↔image cross-attn. for $|i-j| \le w$; shift $s \in \{0, \pm 1\}$ (train)
Physics prior (pvlib, Ineichen)
P^{clr} for all 16 frames (known ahead)
anchor $\hat{P}^{\mathrm{SPM}}_{t+k} = k_t\,P^{\mathrm{clr}}_{t+k}$
availability flags a_P, $a_I \in \{0, 1\}$
history p -- anchor $\mathbf{x}_\tau^P$
target $\mathbf{x}_0^P = (P_{t+k} - \hat{P}^{\mathrm{SPM}}_{t+k})/s_P$
32
64
128
256
128
64
32
$\hat{\mathbf{v}}^P$
PV branch: 1-D convolution + temporal self-attention over all 16 steps, FiLM($\mathbf{e}_\tau$)
projected DDIM (η=0, S=20)
$\hat{\mathbf{x}}_0 = \Pi_C\left(\sqrt{\bar{\alpha}_\tau}\,\mathbf{x}_\tau - \sqrt{1-\bar{\alpha}_\tau}\,\hat{\mathbf{v}}\right)$
generated $\hat{\mathbf{I}}$ (member 1)
observed $\mathbf{I}$
+2 min
+8 min
+16 min
spread cal. s_k (val.)
calibrated PV ensemble, M=30 members (kW)
8
5
2
−14
0
16
sky-image block (H^2 / channels)
PV block (channels)
CA-MMA: cloud-aware RS-PV-MMA
skip (concat.)
physics conditioning
5–95%
median
SPM
obs.